

FORMULAE LIST

Volume of a sphere: $V = \frac{4}{3}\pi r^3$

Volume of a cone: $V = \frac{1}{3}\pi r^2 h$

Volume of a pyramid: $V = \frac{1}{3}Ah$

1. *Loch Lochy*, located in the Scottish Highlands, contains 1.1×10^{12} litres of water. (2)

An Olympic-sized swimming pool contains 2.5×10^6 litres of water.

Calculate how many times an Olympic-sized pool could be filled using the water from Loch Lochy.

2. The value of a house is expected appreciate by 3.4% each year. (3)

The house is valued at £138 000 in April 2026.

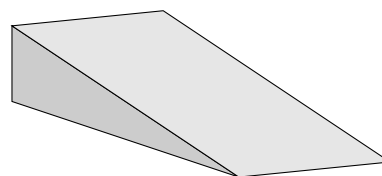
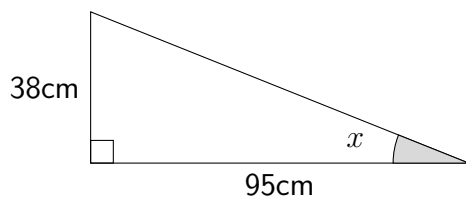
Find the expected value of the house in April 2029.

Give your answer correct to three significant figures.

3. Make p the subject of the formula $t = \frac{p-4}{r}$. (2)

4. A ramp is to be constructed for a skate park as shown in the diagram below. (3)

A side-on view with its height and length is beside a sketch of the completed ramp.



Determine the size of the angle of incline, shown as x on the diagram.

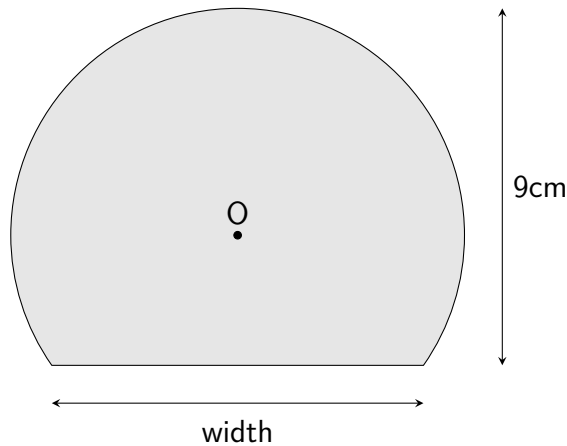
5. Residents of a Scottish island get a 15% discount on the cost of a ferry ticket. (3)

With the residents' discount, the cost of a ticket is £55.08.

Calculate the cost of a ticket without the discount.

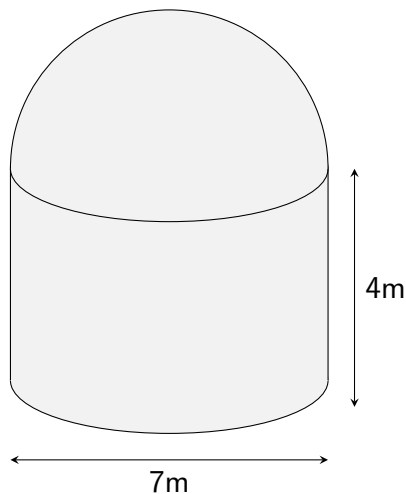
6. Simplify $\left(\frac{4}{5}x^{-3}\right)^2$. (2)

7. The diagram below shows part of a circle, centre O, with diameter 12 centimetres. (4)
The height of the shape is 9 centimetres.



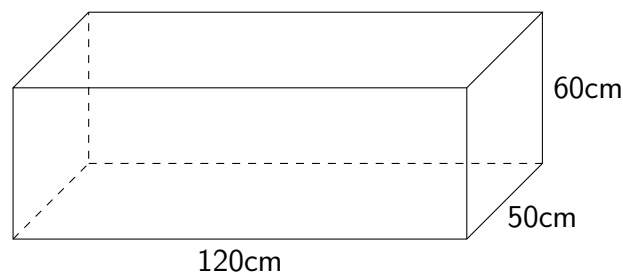
Calculate the width of the base of the shape.

8. An observatory for monitoring the night sky is in the shape of a hemisphere on top of a cylinder. (4)
The height of the **cylindrical part** is 4 metres, and the width of the observatory is 7 metres.



Calculate the volume of the observatory.

9. The diagram below shows a garden storage box in the shape of a cuboid, which measures 50 centimetres by 120 centimetres by 60 centimetres. (4)



A broom handle which is 140cm in length is planned to be stored in the storage box.
Determine whether or not the broom handle can fit inside the box, justifying your answer.

HINTS: Practice Exam A Paper 2

1. Which **operation** lets you calculate how many times one number can go into another?
2. What can you multiply the house's value by to make it **increase** by 3.4%?
3. Start by removing the **denominator** or $r...$
4. If you are getting 15% off, what percentage is **left**? How can we get back to 100%?
5. To find an angle within a right-angled triangle, use **SOH CAH TOA**. Label each side first.
6. How can a fraction be squared? The index law for a **power of a power** says $(x^m)^n = x^{m \times n}$.
7. Start by isolating a **right-angled triangle**. Draw a suitable radius first.
8. The formulae for spheres and cones are given. What is the formula for a **cylinder**? What is the **radius** of each shape?
9. The longest possible distance within a cuboid is from one corner to the opposite corner. Find your notes and examples on **3D Pythagoras**.

ANSWERS: Practice Exam A Paper 2

1. 440 000
2. £153 000
3. $p = rt + 4$
4. 21.8°
5. £64.80
6. $\frac{16}{25}x^{-6}$
7. 10.4cm
8. 243.7m^3
9. $143.2 < 140$ so **the broom handle will fit**